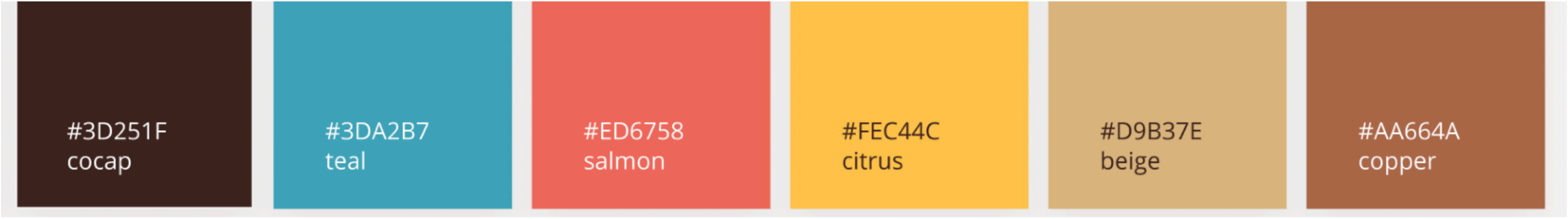


$\pi^\star$ gives mass only to geodesics



solution $\rho_t^\star$ of Benamou-Brenier coincide with



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper





Dynamic Formulation on Path Space

- Dynamic OT over the space of paths:

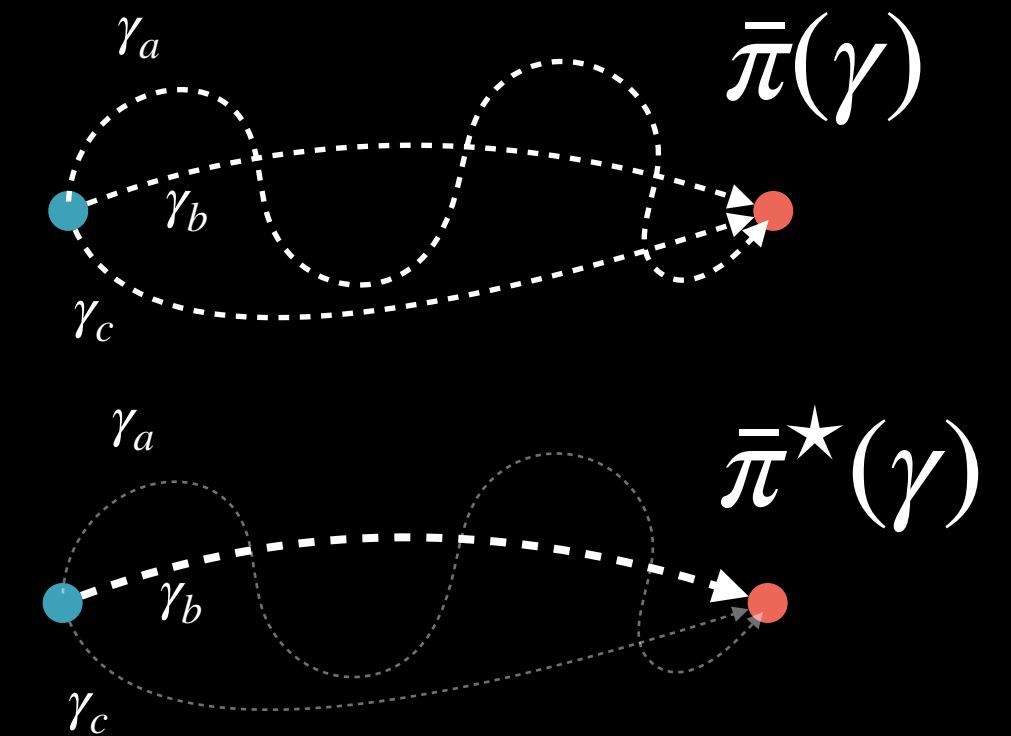
$$W_2^2(\rho_0, \rho_1) = \min_{\bar{\pi} \in \bar{\Pi}} \int_{\bar{X}} L(\gamma)^2 d\bar{\pi}(\gamma)$$

- How does minimizer $\bar{\pi}^\star$ relate to minimizer $\pi^\star$ of static Kantorovich?
- $\bar{\pi}^\star$ gives mass only to geodesics joining pairs (x, y) in proportions prescribed by $\pi^\star$
- For discrete measures, this means:

$$\pi^\star = \sum_{i,j} \pi_{i,j} \delta_{(x_i, y_j)} \quad \text{and} \quad \bar{\pi}^\star = \sum_{i,j} \bar{\pi}_{i,j} \delta_{\gamma_{x_i, y_j}}$$

where γ_{x_i, y_j} is the geodesic between x_i, y_j

- Measures defined by taking ‘snapshots’ of paths at time t , $\rho_t \triangleq (P_t)_\# \bar{\pi}^\star$ coincide with solution $\rho_t^\star$ of Benamou-Brenier



Schrödinger Problem